Fiscal Year:	FY 2011	**Task Last Updated:**	FY 06/01/2011
PI Name:	Hargens, Alan R. Ph.D.		
Project Title:	Rodent spine deconditioning after 30 days of microgravity		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology (2) Neurobiology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
PI Email:	ahargens@ucsd.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	858-534-7837
Organization Name:	University of California, San Diego		
PI Address 1:	Altman Clinical and Translational Research Institute		
PI Address 2:	9452 Medical Center Drive/0863		
PI Web Page:			
City:	La Jolla	**State:**	CA
Zip Code:	92037-0863	**Congressional District:**	52
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2007 Space Flight: Bion-M1 NNH07ZTT001N
Start Date:	08/01/2009	**End Date:**	07/31/2014
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA ARC
Contact Monitor:	Austin, Edward	**Contact Phone:**	650-604-2108
Contact Email:	edward.m.austin@nasa.gov		
Flight Program:	Shuttle		
Flight Assignment:	STS-131 ; Bion NOTE: Extended to 7/31/2014 per A. Chu/ARC (Ed., 4/30/2014)		
Key Personnel Changes/Previous PI:	New Co-I: Koichi Masuda, M.D., Professor, Department of Orthopaedic Surgery, 3525 John Hopkins Court, RM124, San Diego, California 92121. Phone 858-246-0426; koichimasuda@ucsd.edu ; New Co-I: Brinda K. Rana, Ph.D., Assistant Professor, Director, Moores UCSD Cancer Center Sequencing Core. 858-822-4010; bkrana@ucsd.edu		
COI Name (Institution):	Lotz, Jeffrey (University of California, San Francisco) Masuda, Koichi (University of California, San Diego) Rana, Brinda K (UCSD Cancer Center Sequencing Core)		
Grant/Contract No.:	NNX09AP11G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The vertebral bodies and flexible intervertebral discs of the spine are important, weight-bearing tissues that are adapted to gravitational stress. The absence of gravitational axial loads on the body during exposure to microgravity may disrupt normal spine physiology and function. During microgravity exposure, deconditioning of the intervertebral discs and spinal tissues poses a serious risk upon re-exposure to 1-G and exercise. Our recent data in identical twins exposed to 30-days bed rest indicate that compressibility increases in the lumbar spine and its intervertebral discs during 30-days simulated microgravity. Because intervertebral discs become more collapsible during simulated microgravity, the probability of disc prolapse or herniation upon re-exposure to gravity increases. We hypothesize that nucleus pulposus swelling pressures in mouse spines will significantly decrease following 15- and 30-days of microgravity. In addition, we hypothesize that lack of gravitational loading on lumbar discs during microgravity will lead to extracellular matrix damage and significantly reduced disc compression performance following 15- and 30-days of microgravity. To test our hypotheses, we will harvest and test spines from ground-based control mice and flight mice exposed to 15 and 30-days of actual microgravity on STS-131 and Bion-M1. Spines will be frozen and transported to UCSD and UCSF for accepted methods of testing nucleus pulposus swelling pressure, lumbar disc dimensions, proteoglycan content, and spine biomechanics. We will compare the results of the present study to our peer-reviewed and published results for identical twins exposed to 30-days simulated microgravity and control mice exposed to 1-G. Our research will aid understanding of spinal deconditioning during 15- and 30-days microgravity and of the higher incidence of disc prolapse or herniation following re-exposure to 1-G with a long-term view to prevent such spinal deconditioning with exercise or other physiologic countermeasures.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	To our knowledge, this study is the first to examine the effects of 15- and 30-days space flight on spinal deconditioning in mice. The vertebral bodies and flexible intervertebral discs are important, weight-bearing tissues that have adapted to gravitational stress. Our research will aid understanding of spinal deconditioning during 15- and 30-days microgravity and of the higher incidence of disc prolapse or herniation following re-exposure to 1-G with a long-term view to prevent such spinal deconditioning with exercise or other physiologic countermeasures. This research may aid understanding of spinal deconditioning during inactivity such as after spinal cord injury and bed rest in human patients on Earth.
Task Progress:	Objective: The purpose of this study was to analyze the effects of microgravity on intervertebral discs. Also, we hypothesize that there is an increase in water content in the brains of “flight mice” versus the brains of “ground control mice.” Methods: Eight C57BL/6 mice (16-weeks old) were divided into two groups: 1) exposed to microgravity conditions during a 15-day NASA mission, STS-131; and 2) ground-based controls. After animal sacrifice, the caudal intervertebral discs were first imaged by x-ray to calculate disc height and cross-sectional area. Then the C7/C8 motion segments were harvested for biomechanical testing. Discs were subjected to five cycles of compressive creep loading at 0.5 MPa for 20 min with 40 min recovery. The displacement vs. time data were fit to a fluid transport model to calculate tissue permeability (K), time-dependence of annular strain (G), and strain-dependence of nucleus swelling pressure (D). We also studied the effects of 15 days of microgravity exposure on disc structure of flight mice compared with ground control mice. Separately, wet and dry weights of flight and control mice brains were measured. Results: We observed that flight discs were 32% shorter (p=0.012) and had a 70% decrease in D (p=0.02) compared to controls. D was moderately correlated with disc height (R2=0.56, p=0.0013). No statistically significant differences were observed for K and G. In our morphology studies, we documented a loss of disc height during 15 days of microgravity. In particular, a marked disc height loss at the L5/6 level and L6/S1, the lower lumbar level, was found. In our brain studies, a significant difference is opposite to what we hypothesized with the control mice having more brain water than flight mice. Discussion and Conclusions: These differences in D and height are comparable with historical data for mouse caudal discs degenerated by static compressive loading. Consequently, our data suggest that 15-days of microgravity induce intervertebral disc degeneration. Separately, in our morphology studies, we documented a loss of disc height, possibly due to degeneration of the discs, during 15 days of microgravity. In particular, a marked disc height loss at the L5/6 level and L6/S1, the lower lumbar level, was found. This parallels somewhat the high incidence of lumbar disc herniation in astronauts (Johnston et al., 2010). Additionally, decreased disc height at the posterior zone was documented in our study, suggesting increased lordosis. In the MRI analyses, the differences between the control and flight groups were not significant. To assess further the effects of microgravity on the spine, other means of examination (e.g., bone mineral density and trabecular analysis) are currently on-going. Because the anatomical features were not the same between mice and humans (i.e. the meninges is not represented with the mice and we sampled a region rostral of the optic chiasm), the current data do not necessarily contradict the previous human data. Obtaining the meninges and earlier freezing of brain tissues should be considered for future studies. Reference: Johnston, S.L., Campbell, M.R., Scheuring, R., and Feiveson, A.H. 2010. Risk of herniated nucleus pulposus among U.S. astronauts. Aviation, space, and environmental medicine 81:566-574.
Bibliography Type:	Description: (Last Updated: 08/25/2025)
Abstracts for Journals and Proceedings	Bailey JF, Cheng KK, Hargens AR, Masuda K, Lotz JC. "Effects of Microgravity on the Creep Behavior of Murine Intervertebral Discs." Presented at Experimental Biology 2011, Washington, DC, April 9-13, 2011. FASEB Journal 2011 April;25 (Meeting Abstract Supplement):668.1. http://www.fasebj.org/content/vol25/1_MeetingAbstracts/ , Apr-2011
Abstracts for Journals and Proceedings	Bailey JF, Cheng KK, Hargens AR, Masuda K, Lotz JC. "STS-131: Effects of Microgravity on the Creep Behavior of Murine Intervertebral Discs." 18th IAA Humans in Space Symposium, Houston, TX, April 11-15, 2011. 18th IAA Humans in Space Symposium, Houston, TX, April 11-15, 2011. , Apr-2011